🧪 Determination of Chloride Content in a Water Sample (Mohr's Method)

👤 Student Details

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Name: SHUBHAM KUMAR

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Start Experiment

✅ Welcome **SHUBHAM KUMAR** (Roll No. RA2611003010680). Your unknown chloride sample has been assigned. Proceed with Titration 1, then Titration 2.

🎯 AIM

To determine the amount of **chloride ions (Cl^-)** present in the given unknown sample by **argentometric titration (Mohr's method)**

📘 PRINCIPLE

Chloride ions are estimated by titrating the sample, in a **neutral or slightly alkaline medium (pH 6.5 – 10)**, with standard **silver nitrate** solution, using **potassium chromate (K_2CrO_4)** as the indicator.

1. Titration reaction: Ag^+ from the burette reacts with Cl^- in the flask to form a **white precipitate** of silver chloride:

$$\mathrm{Ag^+_{(aq)} + Cl^-_{(aq)} \longrightarrow AgCl_{(s)} \downarrow (\text{white})} \qquad K_{sp} = 1.8 \times 10^{-10}$$

2. End point (indicator reaction): AgCl is less soluble than silver chromate, so it precipitates first. Once all the Cl^- has been removed, the next drop of excess Ag^+ reacts with the chromate indicator to form a **red-brown precipitate** of silver chromate, which marks the end point:

$$\mathrm{2Ag^+_{(aq)} + CrO^{2-}_{4\,(aq)} \longrightarrow Ag_2CrO_{4\,(s)} \downarrow (\text{red-brown})} \qquad K_{sp} = 1.1 \times 10^{-12}$$

Colour change at the end point: Yellow (with white ppt) → **Permanent pale red-brown**

3. Why pH must be 6.5 – 10:

- In **acidic** medium, chromate is converted to dichromate ($2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$), so the end point appears late or not at all.
- In **strongly alkaline** medium, silver is precipitated as AgOH / Ag_2O, which gives a false (high) result.

🧮 CALCULATIONS

- **Titration 1 – Standardization of $AgNO_3$:** $N_{NaCl} \times V_{NaCl} = N_{AgNO_3} \times V_{AgNO_3}$
- **Titration 2 – Chloride in the sample:** $N_{Cl^-} = \dfrac{N_{AgNO_3} \times V_{AgNO_3}}{V_{sample}}$
- **Amount of chloride:** $\mathrm{Cl^-\ (mg/L)} = N_{Cl^-} \times 35.45 \times 1000$

> 👇 Run the cells below to start the **Interactive Titration Bench**: complete Step 1 (preparation), Step 2 (titrant addition) and Step 3 (recording concordant readings).

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Titration 1: Standardization of $AgNO_3$

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🧪 Advanced Interactive Bench: Mohr's Titration Run

Titration 1: Standardization of $AgNO_3$

🧪 Standard NaCl Solution Normality: 0.0200 N

🧪 Step 1: Lab Preparation Checklist

1. Add 10 mL Standard NaCl to Flask ✓ — Add NaCl → 2. Fill the Burette with $AgNO_3$ ✓ — Fill Burette → 3. Add Indicator (K_2CrO_4) ✓ — Add Indicator

💧 Step 2: Titrant Addition (Burette Controls)

$AgNO_3$ (mL): 13.20

+0.05 mL | +0.10 mL | +1.00 mL | Refill Burette

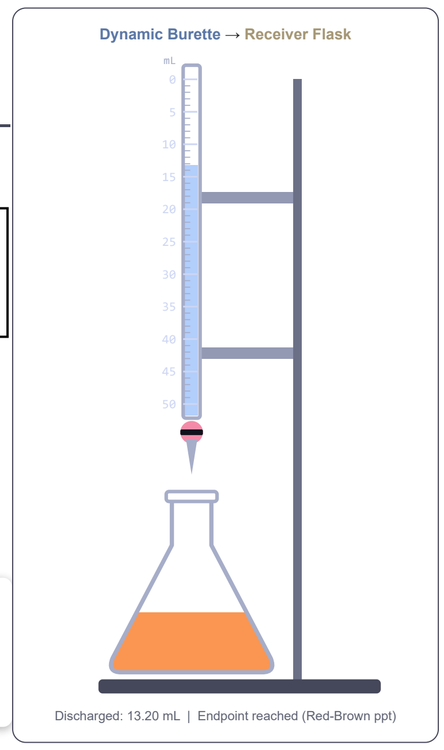
📋 Step 3: Record & Trial Management

Recorded | 🧪 Prepare Next Trial | 🔄 Reset Log

📊 Trial Results Log:

1. Endpoint Volume: **14.00 mL**
2. Endpoint Volume: **13.20 mL**
3. Endpoint Volume: **13.20 mL**

Concordant Volume: 13.20 mL (From trials: 13.20, 13.20)



Discharged: 13.20 mL | Endpoint reached (Red-Brown ppt)

Titration 2: Unknown Sample Chloride Estimation

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Titration 2: Unknown Sample Chloride Estimation

❓ Sample: Unknown Chloride Solution

🧪 Step 1: Lab Preparation Checklist

1. Add 10 mL Unknown Sample to Flask ✓ — Add Sample → 2. Fill the Burette with $AgNO_3$ ✓ — Fill Burette → 3. Add Indicator (K_2CrO_4) ✓ — Add Indicator

💧 Step 2: Titrant Addition (Burette Controls)

$AgNO_3$ (mL): 16.90

+0.05 mL | +0.10 mL | +1.00 mL | Refill Burette

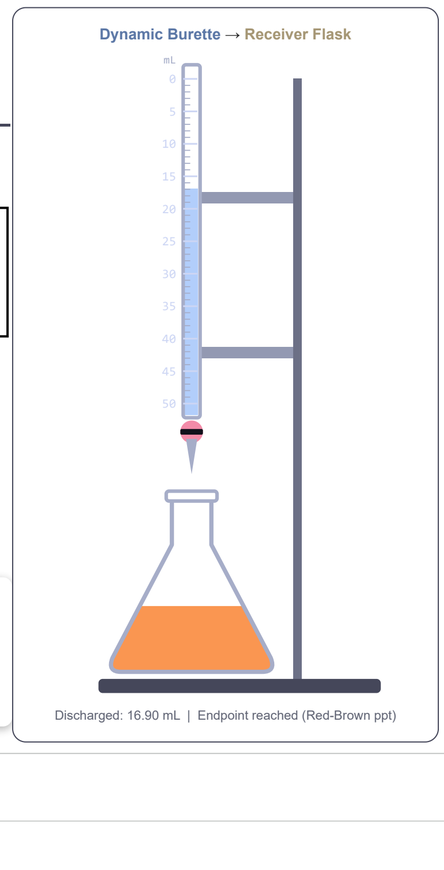
📋 Step 3: Record & Trial Management

Recorded | 🧪 Prepare Next Trial | 🔄 Reset Log

📊 Trial Results Log:

1. Endpoint Volume: **17.00 mL**
2. Endpoint Volume: **16.90 mL**
3. Endpoint Volume: **16.90 mL**

Concordant Volume: 16.90 mL (From trials: 16.90, 16.90)



Discharged: 16.90 mL | Endpoint reached (Red-Brown ppt)

Result: Chloride Content of the Sample

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📋 Result

After recording concordant trials in both titrations, click the button.

Calculate Result

👤 Student

Name: **SHUBHAM KUMAR** | Roll No.: **RA2611003010680**

Titration 1: Standardization of $AgNO_3$

Trial	Burette reading (mL of $AgNO_3$)
Trial 1	14.00
Trial 2	13.20
Trial 3	13.20
Concordant value	**13.20 mL**

$N_{AgNO_3} = (N_{NaCl} \times V_{NaCl}) / V_{AgNO_3} = (0.0200 \times 10.0) / 13.20 =$ **0.0152 N**

Titration 2: Chloride in Unknown Sample

Trial	Burette reading (mL of $AgNO_3$)
Trial 1	17.00
Trial 2	16.90
Trial 3	16.90
Concordant value	**16.90 mL**

$N_{Cl^-} = (N_{AgNO_3} \times V_{AgNO_3}) / V_{sample} = (0.0152 \times 16.90) / 10.0 =$ **0.0256 N**

Chloride (mg/L) = $N_{Cl^-} \times 35.45 \times 1000 = 0.0256 \times 35.45 \times 1000 =$ **907.7 mg/L**

📋 RESULT

The normality of the given $AgNO_3$ solution = **0.0152 N**

The normality of chloride in the given sample = **0.0256 N**

The amount of chloride present in the given sample = **907.7 mg/L**

Actual chloride content of your sample = **901.8 mg/L**

Percentage error = |Experimental − Actual| / Actual × 100 = |907.7 − 901.8| / 901.8 × 100

Percentage error = **0.65 %**